LANGUAGE AS A CLINICAL INTERFACE: NLP-BASED ASSESSMENT, ILLNESS MONITORING, AND RESPONSIBLE AI IN SCHIZOPHRENIA

Start ≈ 0:00 · 41 words

Thank you, Professor Kishimoto. Good afternoon. I am Hironobu Nakamura, a psychiatrist at the Institute of Science Tokyo. Today I will talk about language as a clinical interface: how NLP can help us assess schizophrenia, and how to use it responsibly.

Next: 2 · COI Disclosure

COI DISCLOSURE

Start ≈ 0:25 · 18 words

I have no conflicts of interest to disclose, and the study was approved by the relevant ethics committees.

LANGUAGE: A WINDOW INTO THE MIND

Start ≈ 0:36 · 121 words

Let me start with a simple question: What does language mean to us?

In daily practice, on the ward or in the clinic, our patients' words are our richest source of information. Language is a window into the mind. It helps us recognize symptoms, make diagnoses, and understand our relationships and our humanity.

Yet for more than a century, we have described speech only qualitatively: tangential, impoverished, or loosely associated. Natural language processing now allows us to measure these patterns.

I have been exploring this within the UNDERPIN project, led by Professor Kishimoto.

Today I will cover a brief history, the basic methods, our study, recent trends, and finally the promise and the risks of bringing these tools into the clinic.

Next: 4 · Two Histories Converge

TWO HISTORIES CONVERGE

Start ≈ 1:50 · 96 words

This slide shows how two fields developed in parallel and gradually converged.

In psychiatry, Bleuler described the loosening of associations in 1911, and Andreasen later made thought disorder measurable with a rating scale.

In computing, NLP moved from hand-written rules to statistical methods and latent semantic analysis, then to word embeddings such as Word2Vec in 2013, the Transformer in 2017, and BERT. Now, large language models such as ChatGPT and Claude are part of everyday life. Around 2007, the two lines began to converge, when latent semantic analysis was first used to measure coherence in schizophrenia.

FOUR FAMILIES OF NLP MODELS

Start ≈ 2:48 · 90 words

Next, we need to decide how to measure it. Roughly, there are four families of models.

Counts and rules tell us which words and structures are used. Word embeddings such as Word2Vec measure how close words are in meaning. Contextual models such as BERT capture meaning in context. And large language models can rate and generate text.

Moving to the right, representational power increases, but interpretability decreases. Because our goal was clinical understanding, we deliberately used the first three families, so that every marker can be explained to a clinician.

TRANSITION

Start ≈ 3:43 · 7 words

Now, let’s move on to our study.

INTRODUCTION

Start ≈ 3:47 · 117 words

Language in schizophrenia has long been studied, and NLP now allows us to quantify it, with potential uses in detection, monitoring, and the prediction of onset, prognosis, and relapse.

Previous studies reported reduced syntactic complexity and reduced semantic coherence, as well as poverty of content, lack of cohesion, and changes in pronoun use. Computational models followed: speech graphs for positive thought disorder, semantic density for negative thought disorder, and vector-based models such as latent semantic analysis for coherence.

However, most prior work was limited by a narrow focus, small samples, and single-task designs.

We therefore conducted a large-scale, multi-feature NLP analysis. Our hypothesis was that reduced syntactic complexity and increased semantic proximity are core abnormalities in schizophrenia.

CONCEPTUAL DIAGRAM 1: INTEGRATIVE APPROACH

Start ≈ 4:58 · 91 words

We took an integrative approach to capture linguistic abnormalities across multiple levels of language, combining several NLP methods.

At the morphological level: word counts, parts of speech, and pronoun use.

At the syntactic level: sentence complexity, such as tree depth and clause nesting depth.

At the semantic level: similarity between words, using static word embeddings, and the density of semantic networks.

At the contextual level: sentence similarity, coherence, and continuity across discourse.

Prior studies usually examined these levels separately. We bring them together in one comprehensive analysis of a large sample.

Next: 9 · Conceptual Diagram 2 : Past and Present

CONCEPTUAL DIAGRAM 2 : PAST AND PRESENT

Start ≈ 5:54 · 94 words

This study also connects past and present. Linguistic abnormalities in schizophrenia have long been described through classical psychopathology and linguistics. We treated those classical descriptions as hypotheses and tested them with NLP and quantitative linguistics.

For example, poverty of thought can be measured through vocabulary, fillers, and perseveration; neologisms as words absent from a dictionary; incoherence as syntactic abnormality; and loosening of associations as abnormal semantic similarity between words.

This is a transition from descriptive interpretation to quantitative analysis. Our goal is to bridge classical insight and modern computation, combining hypothesis-driven and data-driven approaches.

MATERIAL & METHOD: PARTICIPANTS

Start ≈ 6:51 · 42 words

The study was part of the UNDERPIN project. We included 104 patients with schizophrenia and 101 healthy controls from seven hospitals and three clinics in Japan, between 2018 and 2024.

All participants were adults fluent in Japanese and gave written informed consent.

Each participant had two to five visits, at least one month apart, and symptoms were rated with the PANSS.

Each interview included three tasks: free conversation, picture description with standardized images, and storytelling, retelling Cinderella or the Japanese folktale Kaguya-Hime. All sessions were recorded with headset microphones.

METHOD: SPEECH ANALYSES

Start ≈ 7:45 · 41 words

Speech was transcribed, and only the participant's speech was analyzed. We extracted 76 features at the four levels I described: the GiNZA parser for part-of-speech tagging, named entities, and syntax, Word2Vec and TF-IDF for semantics, and SentenceBERT for coherence and perseveration.

METHODS: ANALYSIS PIPELINE

We conducted statistical analyses according to this pipeline, after calculating the NLP features.

For feature processing, we processed 76 NLP features for each task and each visit. To adjust for differences in speech length, features were expressed as ratios where appropriate, and then z-standardized using parameters from the training set.

For exploratory analyses, we compared patients and healthy controls using univariate generalized estimating equations. These results were used for description only, not for final feature selection.

To reduce dimensionality, we conducted exploratory factor analysis. From each factor, we selected a representative feature based on the largest absolute loading, while avoiding multicollinearity.

For validation, we fitted task-specific GEEs. The outcome was schizophrenia versus control, and the predictors were the representative features, with age and sex as covariates.

Finally, we evaluated predictive performance on the held-out test set, using AUC, sensitivity, and specificity. All preprocessing and feature selection were confined to the training set to prevent information leakage.

RESULTS: DEMOGRAPHICS

Start ≈ 9:45 · 29 words

The groups were similar in age and sex. Patients had higher PANSS scores, mostly in the mild-to-moderate range, and fewer years of education. Almost everyone completed all three tasks.

METHODS: ANALYSIS PIPELINE

Start ≈ 10:03 · 4 words

First, the exploratory comparisons.

RESULTS: EXPLORATORY ANALYSES

Start ≈ 10:05 · 33 words

In free conversation, 35 of the 76 features differed after FDR correction, across all levels of language. For example, patients used more person-related named entities and fewer adverbs. These results are descriptive only.

RESULTS: EXPLORATORY ANALYSES (SEMANTIC NETWORK)

Start ≈ 10:26 · 24 words

In the semantic network analysis, the clustering coefficient was higher in patients, and it correlated with PANSS negative symptoms and with fluency of speech.

SEMANTIC NETWORK ANALYSIS USING A TF-IDF FRAMEWORK

Start ≈ 10:40 · 59 words

Let me take a moment to explain semantic network analysis briefly. For each participant, we took the most characteristic words using TF-IDF, and connected words that were close in meaning using Word2Vec.

Then we measured the shape of the network: clustering, diameter, distance, and centrality. By eye, patient and control graphs look similar, so we rely on these numbers.

SUMMARY OF RESULTS: SEMANTIC NETWORK ANALYSIS

Start ≈ 11:16 · 93 words

Here is the summary. In schizophrenia, the clustering coefficient and density increased, while average distance, diameter, and closeness centrality decreased. The asterisks mark the significant features.

In other words, semantic connections are overly tight. Patients tend to stay within close semantic neighborhoods, and distant ideas are harder to reach. This over-connectedness does not make information transfer more efficient.

On the right, I show schematic graphs. They may not be very clear, but they illustrate the idea that in schizophrenia, local clusters are denser, yet connections to more distant nodes are not easily accessible.

METHODS: ANALYSIS PIPELINE

Start ≈ 12:13 · 18 words

With so many differences, we asked which dimensions are truly essential. To find out, we used factor analysis.

RESULT: FACTOR ANALYSIS & GEE (TASK 1)

Start ≈ 12:24 · 120 words

In free conversation, factor analysis gave a stable nine-factor solution, and we kept one representative feature per factor, with low collinearity.

In the multivariable model, adjusted for age and sex, three features remained significant. First, higher mean pairwise word similarity: one standard deviation more was associated with about 2.3-fold higher odds of schizophrenia. Second and third, lower ratios of case particles and adverbs.

In Japanese, case particles such as ga, o, and ni mark who did what to whom, and adverbs add manner, degree, and the speaker’s stance. Among the major parts of speech, only adverbs were reduced.

In the independent test set, the model reached an AUC of 0.87, with a sensitivity of 0.76 and a specificity of 0.90.

RESULT: COMPARISON OF TASKS

Start ≈ 13:37 · 80 words

We then applied the same pipeline to storytelling and picture description.

Adverbial features were significant in every task, and case particles were reduced regardless of task. Semantic-network and discourse features also contributed. Perseveration was strongly elevated in storytelling, mirroring the clinical phenomenon.

Sensitivity analyses showed that the adverb effect was strongest in free conversation, while case particles were reduced in every task. So case-particle reduction looks like a stable, task-general simplification, while reduced modification is most visible in spontaneous speech.

Taken together, we propose a tripartite linguistic phenotype of schizophrenia.

First, reduced morphosyntactic explicitness: fewer case particles, so less overt marking of who did what to whom.

Second, semantic narrowing: words drawn from close neighborhoods.

Third, weakened contextual modulation: fewer adverbs, so less manner, degree, and nuance.

These dimensions are separable but interacting, and they go beyond the positive-negative dichotomy of thought disorder. Similar findings in Turkish, another agglutinative language, suggest shared vulnerabilities shaped by language type, and the semantic findings parallel neuroimaging studies of network integration.

Clinically, these markers are simple, interpretable, and related to symptom severity. But they remain hypotheses, to be tested with cognitive and functional measures.

LIMITATIONS

Start ≈ 15:33 · 42 words

There are four main limitations. First, we had no clinical high-risk group, so we cannot predict onset. Second, most patients had mild to moderate illness. Third, all data were in Japanese, so cross-linguistic validation is needed. Fourth, we analyzed spoken language only.

TRANSITION

Start ≈ 15:58 · 13 words

Now, let’s move on to recent trends, and to the promise and peril.

NLP × PSYCHIATRY: RECENT TRENDS

Start ≈ 16:06 · 58 words

Let me place our study in recent research. Early studies showed that speech can predict psychosis onset. Research then turned to language dimensions and mechanisms, and large language models brought new opportunities and risks.

Today, the field is moving toward clinical use, with a focus on relapse prevention. Our next step is longitudinal monitoring with these interpretable features.

PROMISE AND PERIL

Start ≈ 16:41 · 154 words

I began by describing language as a window into the mind. That image captures both the promise and the risks of NLP.

On the promise side, these tools could support objective severity assessment, earlier warning of relapse, and tracking of treatment response. They could enable monitoring between visits, reduce variability between raters, and support screening at scale.

But speech can reveal a person's identity and deeply private experiences. Monitoring may feel like surveillance and erode trust, especially without meaningful consent. Models can be biased across languages, dialects, education, and culture. Large language models can be hard to interpret and may generate false information. There are also risks of overdiagnosis, labelling, and over-trust in automated judgments, and generative AI may echo or reinforce delusional beliefs.

Language data are intimate data. The same window that helps us understand a person can also leave them exposed. Our responsibility is to improve care while protecting privacy and autonomy.

TOWARD RESPONSIBLE USE

Start ≈ 18:15 · 74 words

Responsible use rests on three pillars: evidence, trust, and clinical integration.

First, evidence. We need external and cross-linguistic validation, and longitudinal, prospective studies.

Second, trust. Privacy must be built into the design, including de-identification, and the tools must be transparent and explainable.

Third, clinical integration. Clinicians stay involved, and high-stakes decisions such as diagnosis remain in human hands.

The goal is AI that supports and deepens clinical understanding, while clinicians keep responsibility for care.

CONCLUSION: TAKE-HOME MESSAGES

Start ≈ 19:00 · 58 words

To conclude, three messages.

First, language is a clinical interface, and NLP can turn it into measurable, interpretable markers.

Second, in Japanese speech, schizophrenia shows a tripartite phenotype: fewer case particles, semantic narrowing, and fewer adverbs, consistent across tasks.

Third, AI should augment, not replace, clinical understanding, the therapeutic relationship, and human responsibility.

Thank you for your attention.